SE 2228 Analysis & Design of Algorithms
Prof. Ahmet KOLTUKSUZ, Ph. D.
Ras Mert Hazar HASGÜL, Ms in Comp. Eng., Hatice OBUZ, Ms in Comp. Eng
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Student Id:_____ **Name & Last Name:**_____

Section Number:____

Notes:

1. **Duration is two hours.**
2. All course materials are closed.
3. No cell phones, no computers of any kind!
4. **Collaboration is strictly and positively prohibited.**

QUESTIONS

Q.1(5 points) Find the value returned by the following function, express your answer as a function of n, and determine the worst-case running time using the Big Oh notation.

```
function mind-boggling(int n)
{
   int i, j, k, x= 0, n=5;

        for(i= 1; i ≤ n – 1; i++)
                for(j= i + 1; j ≤ n; j++)
                        for k=j; k ≤ j; k++)
                                x= x + 1;
        return(x);
}
```

Answer #1:

$O(n^2)$

first loop i=1 to n-1 → n times

second loop j=i+1 to n → n times

third loop runs only once for j thus it is 1

$O(n * n + 1) = O(n^2+1) = O(n^2)$

x=10

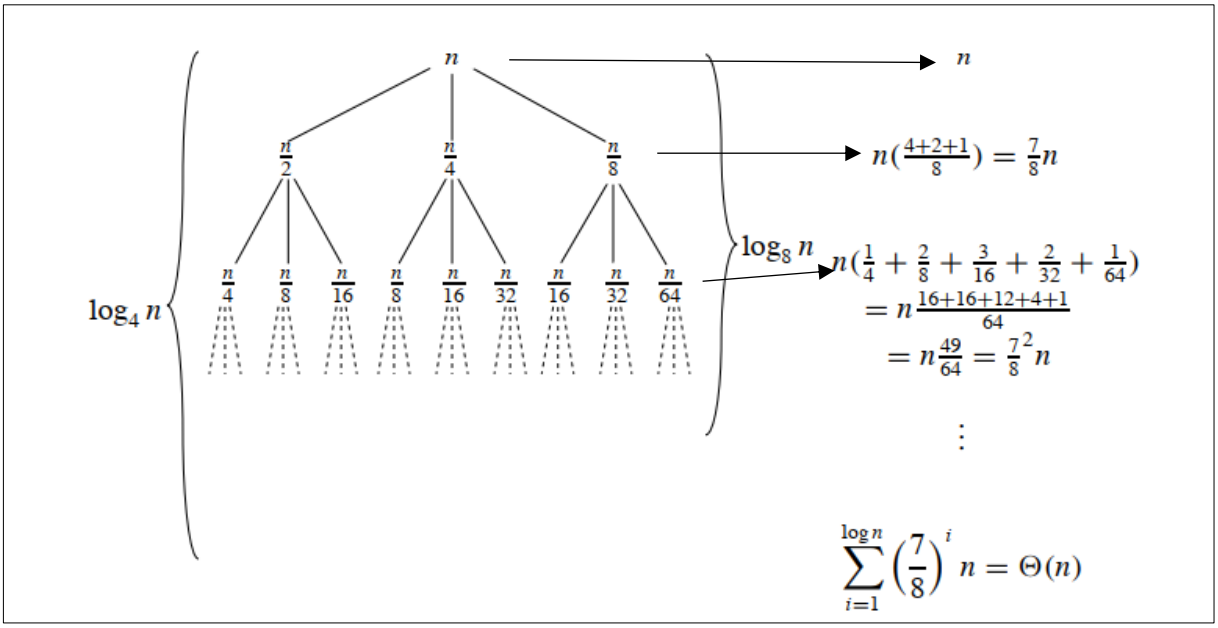
Q.2 (5 points) $\sum_{i=1}^{n} \log_2 4^i = ?$

Answer #2: $\sum_{i=1}^{n} \log_2 4^i = \sum i \,.\, \log_2 4 = 2\sum i \approx n(n+1) = n^2 + n$

Q.3 (22 points) For $T(n) = T(n/2) + T(n/4) + T(n/8) + n$

Provide the recursion tree and the worst-case complexity.

Answer #3:



$$
\begin{aligned}
T(n) &= T(n/2) + T(n/4) + T(n/8) + n \\
&\le cn/2 + cn/4 + cn/8 + n \\
&= 7cn/8 + n \\
&= (1 + 7c/8)n \\
&\le cn \qquad \text{if } c \ge 8 .
\end{aligned}
$$

Therefore, $T(n) = O(n)$.

Answer #3 O(n)

Q.4 (10 points) Find the run time function [T(n)] and complexity of the following code segment:

```
for (i = 1; i <= n*n; i++)
    for (j = 1; j <= 5*n; j++) count++;
        for (i = 1; i <= 3*n; i++) count++;
```

Answer#4:

T(n) =$5n^3$+3n = O(n^3), the count++ gets called $5n^3$+3n times

Q.5 (18 points) Answer the questions below

Answer #5:

a. What are the limitations of Dijkstra's shortest path algorithm?

It won't work for the unweighted graphs.
It calculates the minimum distances from the source vertex to all other vertices. Not from every vertex to the other vertices.

b. Under what circumstances do we choose Quicksort over Mergesort, although they both have the same complexity?

Quicksort sorts in place, thus doesn't need additional memory; however, Mergesort doesn't sort in place and uses additional storage during merges.

c. Define the complexity of an algorithm.

For a given algorithm, the resources required for its execution determine its complexity.

d. Define the complexity of a problem.

The complexity of a problem is the complexity of the algorithm that solves it.

e. When should we not use the Brute-force methodology?

When there is an optional algorithm to solve the problem WITHOUT testing all the possible solution candidates.

f. Other than a prerequisite sorting for Kruskal's and their complexities, what is the main difference between Kruskal's and Prim's MST algorithms?

Kruskal's algorithm maintains a forest that grows until it forms a spanning tree. The alternative is to keep just one tree and grow it until it spans all the nodes, as in Prim's algorithm.

Q.6 (40 points) CODING PROBLEM: There is a company that has a very creative way of managing its accouınts. Every time they want to write down an account number, they peculiarly shuffle its digits: They alternatively write one digit from the front of the number and one digit from the back, and then the second digit from the front and the second from the back, and so on until the length of the shuffled number is the same as that of the original. Now, as the responsible Computer/Software Engineer of this company, you are to write a function like that of an **int solution(int A)** that, given a positive integer A, returns its shuffled representation. **For example, given A=123456, the function should return 162534. And for a given A=130, the function returns 103.** Also, assume that **A** is an integer within the range of **[0…100,000,000].** Now,

a. **(15 points)** Design the solution algorithm most efficiently and correctly, using pseudocode or a flowchart.
b. **(15 points)** Provide the code in C/C++ language.
c. **(10 credits)** Perform a code complexity analysis.

Answer #6:

i. Design

```
begin
        declare number, new-number, string, new-string
        get the number
        convert number to a string
        new-string = manipulate-string(string)
        convert new-string to new-number
        print new-number
        record new-number
end
begin manipulate_string(str)
        get the length of str
        while (length !=0) do manipulation on str as defined
        return (str)
end manipulate string
```

ii. The exemplary code

```
#include<stdio.h>
#include<stdlib.h>

int solution(int A) {
        int tmp = A;
        int size = 0;
        while(tmp != 0) {
                tmp = tmp / 10;
                size++;
        }

        char *str = (char *) malloc(size);
        tmp = A;
        int i = size - 1;
        while(tmp != 0){
                int rem = tmp % 10;
                str[i] = rem + '0';
                i--;
                tmp = tmp / 10;
        }

        int k = size-1;
        int l = 0;
        int j = 0;
```

```
	char *new = (char *)malloc(size);
	while(l<size) {
		new[l] = str[j];
		new[l+1] = str[k];
		j = j+1;
		k = k-1;
		l = l + 2;
	}

	if(size % 2 == 1) {
		new[size] = '\0';
	}

	l = 0;
	int result = 0;

	while(l < size) {
		result = result * 10 + (new[l] - '0');
		l++;
	}

	return result;
}


int main() {

	int A = 1346527;
	int result = solution(A);
	printf("result : %d\n", result);
	return 0;
}
```

iii. The complexity Analysis

Complexity of the solution function is number of digits variable A. Because main loop count for this number.